

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                    |                 |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------|
| <b>Department</b>                        | FAST School of Computing (FSC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Semester</b>    | FALL 2024       |
| <b>Course Title</b>                      | Applied Physics BS(CS) BS(DS) & BS(SE)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Course Code</b> | NS1001          |
| <b>Pre-requisite</b>                     | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Credit Hrs.</b> | 3               |
| <b>Course Moderator</b>                  | Dr. Saman Shahid ( <a href="mailto:saman.shahid@lhr.nu.edu.pk">saman.shahid@lhr.nu.edu.pk</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Semester</b>    | 1 <sup>st</sup> |
| <b>Course Objectives:</b>                | The Applied Physics course is aimed to introduce vector algebra, Newton's law to solve two and three dimensional systems, forces and objects in motion. It is also focused on evaluating simple harmonic motion (SHM), oscillations and waves. The last part of the course is designed for the learning of electricity & magnetism which includes Coulomb's law, Gauss's law, systems of capacitance, Ohm's law, Hall's effect, magnetic forces, current-carrying magnetism, and Ampere's law. The course is a pre-requisite of "Digital Logic Design".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    |                 |
| <b>Course Learning Objectives (CLOs)</b> | <p>At the end of the course, the students will be able:</p> <ol style="list-style-type: none"> <li>1. Find position, displacement, velocity, acceleration in 1, 2 &amp; 3 dimensions in numerical problems or MATLAB simulation code/programming.</li> <li>2. Learn projectile motion with the application of vector analysis to calculate horizontal/vertical motions, equation of the path and horizontal range to apply in numerical problems or MATLAB simulation code/programming.</li> <li>3. Apply Newton's Laws along with vector notations to evaluate different types of forces: gravitational/weight/normal/tension/friction to apply in numerical problems or MATLAB simulation code/programming.</li> <li>4. Verify SHM in learning different oscillations (simple, angular, uniform circular motion) for different pendulums/oscillators (torsional, simple).</li> <li>5. Learn Different Types of Waves (Transverse &amp; Longitudinal), Sinusoidal Waves and their respective parameters: Wavelength, Frequency, Angular Frequency, Wave number, Speed of wave.</li> <li>6. To understand electric charge, electric current, resistance, resistivity and electric field with different applications through associated laws (i.e., Ohm's Law, Coulomb's law &amp; Gauss' Law) and implement them to calculate related physical quantities in numerical problems or MATLAB simulation code/programming.</li> <li>7. To understand different types (parallel plate, cylindrical, spherical) &amp; combinations (parallel/series) of capacitances to calculate capacitances along with the other associated physical quantities (e.g. potential difference) in numerical problems.</li> <li>8. To understand magnetic fields &amp; magnetic forces, their application as current carrying wire, Hall's effect and in circulating charges to calculate related physical quantities to solve numerical problems or MATLAB simulation codes.</li> <li>9. To understand magnetic fields generated due to currents by Ampere's law to calculate magnetic fields due to different conditions and geometries (e.g. Solenoid and Toroid) and calculate related physical quantities to apply in numerical problems or MATLAB simulation.</li> </ol> |                    |                 |

|                     |                  |                                                                                                     |
|---------------------|------------------|-----------------------------------------------------------------------------------------------------|
| <b>Text Book(s)</b> | <b>Title</b>     | <b>Halliday &amp; Resnick Fundamentals of Physics (Extended 11th Edition)</b>                       |
|                     | <b>Author(s)</b> | Jearl Walker                                                                                        |
|                     | <b>Publisher</b> | © 2013 by John Wiley & Sons Inc.                                                                    |
| <b>Ref. Book(s)</b> | <b>Title</b>     | <b>Physics for Scientists and Engineers with Modern Physics (6th Edition)</b>                       |
|                     | <b>Author(s)</b> | Raymond A. Serway & John W. Jewett                                                                  |
|                     | <b>Publisher</b> | © 2004 Thomson books/cole US                                                                        |
|                     | <b>Title</b>     | <b>Physics for Scientists and Engineers (6th Edition)</b>                                           |
|                     | <b>Author(s)</b> | Paul A Tipler and Gene Mosca                                                                        |
|                     | <b>Publisher</b> | W.H. Freeman and Company                                                                            |
|                     | <b>Title</b>     | <b>Physics for Scientists and Engineers (3<sup>rd</sup> Edition)</b>                                |
|                     | <b>Author(s)</b> | Fishbane, Gasiorowicz, Thornton                                                                     |
|                     | <b>Publisher</b> | Pearson Prentice Hall                                                                               |
|                     | <b>Title</b>     | <b>Physics for Engineers &amp; Scientists (3<sup>rd</sup> Edition Extended)</b>                     |
|                     | <b>Author(s)</b> | Hans C. Ohanian and John T. Markert                                                                 |
|                     | <b>Publisher</b> | W. W. Norton & Company New York. London                                                             |
|                     | <b>Title</b>     | <b>MATLAB –A practical introduction to programming and problem solving (3<sup>rd</sup> Edition)</b> |
|                     | <b>Author(s)</b> | Stormy Attaway                                                                                      |
|                     | <b>Publisher</b> | Elsevier 2013                                                                                       |
|                     | <b>Title</b>     | <b>Programming with MATLAB for Scientists</b>                                                       |
|                     | <b>Author(s)</b> | Eugeniy E. Mikhailov                                                                                |
|                     | <b>Publisher</b> | CRC Press Taylor & Francis 2017                                                                     |
|                     | <b>Title</b>     | <b>Introduction to MATLAB for Engineers (3<sup>rd</sup> Edition)</b>                                |
|                     | <b>Author(s)</b> | William J. Palm III                                                                                 |
|                     | <b>Publisher</b> | The McGraw-Hill Companies                                                                           |



| Week                | Course Contents/Topics                                                                                                                                                        | Chapter |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 01                  | Adding Vectors, Components of Vectors, Unit Vectors, Vector & Scalar Products, Position & Displacement (2/3 dimensions), Numerical Problems & MATLAB Implications             | 03, 04  |
| 02                  | Average/Instantaneous Velocity/Acceleration, Uniform Circular Motion, Numerical Problems & MATLAB Implications                                                                | 04      |
| 03                  | Projectile Motion, horizontal/vertical motions, equation of the path, max. height, time of flight and horizontal range, Numerical Problems & MATLAB Implications              | 04      |
| 04                  | Newton Laws of Motion, Forces (1D/2D): Gravitational, Friction, Tension, Weight, Numerical Problems & MATLAB Implications                                                     | 05      |
| <b>MIDTERM – I</b>  |                                                                                                                                                                               |         |
| 05                  | Simple Harmonic Motion, the Force Law for SHM, Angular SHM                                                                                                                    | 15      |
| 06                  | Simple Pendulum, Circular Motion & SHM, Numerical Problems & MATLAB Implications                                                                                              | 15      |
| 07                  | Types of Waves, Sinusoidal Waves, Wavelength and Frequency and Wave Speed.                                                                                                    | 16      |
| 08                  | Coulomb's Law, Charge Quantization & Conservation, Electric Field, Electric Field Due To Point Charge and Dipole, Numerical Problems & MATLAB Implications                    | 21, 22  |
| 09                  | Gauss' Law, Flux, Flux Of Electric Field, Gauss's Law, Equivalency of Gauss's Law And Coulombs' Law                                                                           | 23      |
| 10                  | Cylindrical Symmetry, Planar Symmetry, Spherical Symmetry, Numerical Problems & MATLAB Implications                                                                           | 23      |
| <b>MIDTERM - II</b> |                                                                                                                                                                               |         |
| 11                  | Capacitance, Parallel Plate, Cylindrical & Spherical Capacitors, Capacitors In Parallel And In Series, Numerical Problems & MATLAB Implications                               | 25      |
| 12                  | Electric Current, Current Density and Drift Speed, Resistance & Resistivity, Ohm's Law, Numerical Problems & MATLAB Implications                                              | 26      |
| 13                  | Magnetic Fields And Field Lines, Crossed Fields: Hall Effect, Circulating Charge Particles, Magnetic Force On Current Carrying Wire, Numerical Problems & MATLAB Implications | 28      |
| 14                  | Magnetic Field Due To Current, Ampere's Law, Magnetic Field Inside/Outside Wire, Solenoids & Toroids & Between two Parallel Wires, Numerical Problems & MATLAB Implications   | 29      |

**Teaching Methodology:**

Lecturing, MATLAB laboratory practice, Assignments, Quizzes, Discussion, Midterm exams, Presentations, Simulations etc..

**Course Assessment:**

| Assessment Tools             | Weightage    |
|------------------------------|--------------|
| Quizzes (5), Assignments (5) | 10%+10% =20% |
| Midterms (I+II)              | 15%+15%=30%  |
| Final Exam                   | 50%          |

**Important Instructions:**

- You may have to secure at least 50% marks to pass the course.
- Plagiarism is not tolerable in any of its form; minimum penalty would be an 'F' grade in the course without prior warning.

**Grading Criteria:**

Relative Grading Scheme may be used for the course evaluation.